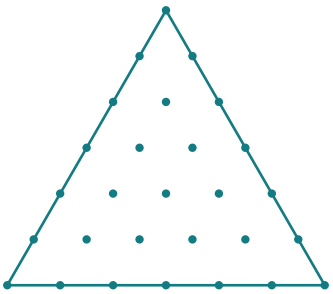
PROBABILITY AND MATHEMATICAL MODELLING

Discrete Information Geometry

Finite Differences, Probability, and Information Flows

Chapter 3



A course in ten chapters

https://bobovski66.github.io/IntroDiscreteInfoGeom/

CHAPTER 3

Discrete gradients on the probability simplex

Suppose a campus survey records three ways of reaching class: walking, cycling, and taking a bus. A distribution such as $(1/2, 1/3, 1/6)$ says that half the responses are in the first category, a third in the second, and a sixth in the third. Changing one share forces at least one other share to change. The three numbers describe a state with two independent degrees of freedom.

Chapter 2 studied motion on a probability interval. With three categories, the state space becomes a triangle. We will use a finite grid of possible survey compositions to develop directional differences, gradients, circulation, and divergence. Each operation will have a concrete meaning in terms of changing one response or moving a quantity along an edge.

3.1 Why probability states form a triangle

Write $p = (p_A, p_B, p_C)$, with nonnegative coordinates adding to one. Knowing p_A and p_B determines $p_C = 1 - p_A - p_B$. The permitted pairs satisfy $p_A \geq 0$, $p_B \geq 0$, and $p_A + p_B \leq 1$.

It is often more symmetric to place pure states at the vertices of an equilateral triangle:

$$A = (0, 0), \qquad B = (1, 0), \qquad C = (1/2, \sqrt{3}/2).$$

The **barycentric representation** places the distribution at the weighted point

$$X = p_A A + p_B B + p_C C = \left(p_B + \frac{p_C}{2}, \frac{\sqrt{3}}{2} p_C \right).$$

The square root appears because an equilateral triangle of side one has height $\sqrt{3}/2$. It only sets the picture's proportions.

At vertex A, all the probability belongs to A. On edge AB, the C coordinate is zero. The center corresponds to $(1/3, 1/3, 1/3)$. Lines parallel to edge AB have constant height, hence constant p_C.

Example 3.1 (Locating and reading a mixture)**.** For $p = (1/2, 1/3, 1/6)$, the plotted point is $(5/12, \sqrt{3}/12)$. Its height is one sixth of the triangle's total height, matching $p_C = 1/6$.
Conversely, if a plotted point has coordinates (u, v), solve the plotting equations:

$$p_C = \frac{2v}{\sqrt{3}}, \qquad p_B = u - \frac{v}{\sqrt{3}}, \qquad p_A = 1 - u - \frac{v}{\sqrt{3}}.$$

At $(u, v) = (1/2, \sqrt{3}/6)$, these give three equal probabilities.

The set of probability vectors on k labels is called the **probability simplex**. Two labels give an interval, three give a triangle, and four give a tetrahedron. The definition is always the same: nonnegative entries with total one. We can calculate with more labels even when we cannot draw their simplex.

3.2 Fixing a number of tokens makes a finite lattice

Imagine a survey of exactly N responses. Counts (a, b, c) satisfy

$$a, b, c \text{ are nonnegative integers}, \qquad a + b + c = N.$$

Their empirical proportions are $p = (a/N, b/N, c/N)$. The possible points form a triangular **lattice**, a finite grid inside the simplex. We denote the set by $\mathcal{P}_N$.

For $N = 4$, group the triples by their C count:

Count c	Possible pairs (a, b)	Number
0	$(4,0), (3,1), (2,2), (1,3), (0,4)$	5
1	$(3,0), (2,1), (1,2), (0,3)$	4
2	$(2,0), (1,1), (0,2)$	3
3	$(1,0), (0,1)$	2
4	$(0,0)$	1

The total is fifteen. In general, fixing c leaves $N - c + 1$ choices, so the number of states is

$$1 + 2 + \cdots + (N + 1) = \frac{(N + 1)(N + 2)}{2}.$$

Increasing N increases the resolution: one response is a smaller fraction of the whole.

A **neighboring state** is obtained by transferring one token from one category to another. From $(2, 1, 1)$, moving a token from C to A gives $(3, 1, 0)$. The probability displacement is $(1/4, 0, -1/4)$. Every transfer preserves the total count and the total probability.

At $(2, 1, 1)$ all three categories can donate, so there are six neighbors. At $(4, 0, 0)$ only A can donate, giving two neighbors. Boundary restrictions are part of the geometry, not a failure of the formula.

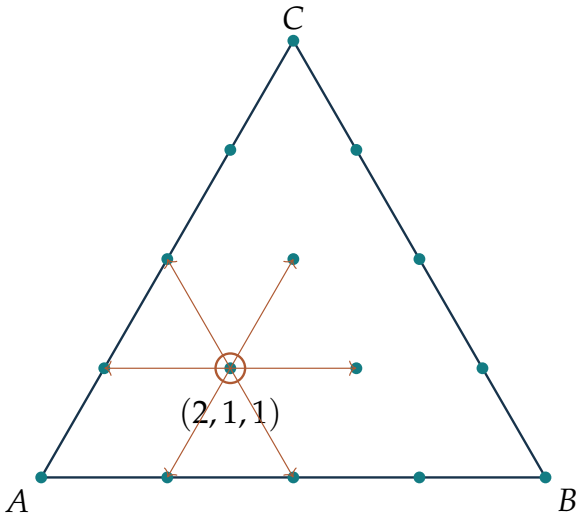


Figure 3.1. The denominator-four token lattice. Arrows show the six one-token moves from counts (2,1,1).

These tokens represent units of composition. A token transfer through probability space need not describe a person physically traveling between locations. Chapter 4 will introduce another graph whose vertices are physical or categorical locations. Keeping these two meanings of a vertex separate will help.

The lattice is also a graph: its count states are vertices and its legal transfers are edges. A drawing of the triangle helps us see the graph, but the definition does not depend on the drawing. A computer can store each triple and list its legal neighbors directly.

To generate those neighbors, choose a donor category with positive count and a different receiver. Subtract one from the donor and add one to the receiver. There are six ordered donor-receiver pairs for three categories, but some are forbidden at the boundary. This procedure gives both an exact state space and an exact rule for adjacency.

The number of interior states can be counted separately. A strictly positive triple has at least one token in each category. Reserve those three tokens; the remaining $N - 3$ are distributed freely. For $N \geq 3$, there are $(N-1)(N-2)/2$ interior states. Subtracting from the full count gives $3N$ boundary states. Vertices belong to two triangle edges in the picture but are counted only once in this total.

For $N = 4$, there are three interior states and twelve boundary states. Thus boundaries are a large part of a coarse lattice. Ignoring illegal moves would change the model substantially, rather than introduce only a small numerical error.

3.3 A directional difference compares two permitted states

A **scalar field** assigns one number to every state. For a first example, consider

$$C(p) = p_A^2 + p_B^2 + p_C^2.$$

This concentration score is one at a pure state and one third at the balanced state. Larger values indicate a more concentrated composition.

Let $e_A = (1,0,0)$, $e_B = (0,1,0)$, and $e_C = (0,0,1)$. These are unit coordinate vectors. A transfer from C to A changes p by $(e_A - e_C)/N$. Write

$$T_{A\leftarrow C}p = p + \frac{e_A - e_C}{N}.$$

The arrow names the receiving and donating categories. The move is legal when $p_C \geq 1/N$.

For any field F, define its **directional difference**

$$\boxed{D_{A\leftarrow C}F(p) = F(T_{A\leftarrow C}p) - F(p).}$$

First evaluate the field after the move, then subtract its value before the move. This is the same operation as a first difference, now taken in a specified direction through a state space.

Example 3.2 (Two directions from the same composition)**.** At counts $(2,1,1)$ with $N = 4$, we have $p = (1/2, 1/4, 1/4)$ and $C(p) = 3/8$.
Moving C to A gives $(3/4, 1/4, 0)$, with concentration $5/8$. The directional difference is $1/4$.
Moving A to B gives $(1/4, 1/2, 1/4)$, again with concentration $3/8$. That directional difference is zero. The state changed, but this field assigns the same number to the two states.

Our D records the total change caused by one token. A difference quotient per unit of transferred probability would divide by $1/N$ and equal $ND_{A\leftarrow C}F$. A quotient per unit of ordinary Euclidean displacement would use the step length $\sqrt{2}/N$. Stating the convention prevents factors from entering unnoticed. We use unscaled edge differences throughout the graph calculus.

We can derive concentration changes for every transfer, rather than calculate a new square sum each time. Let $h = 1/N$, and transfer from category j to category i. Only those two terms change:

$$\begin{aligned} C(p + h(e_i - e_j)) - C(p) &= (p_i + h)^2 - p_i^2 + (p_j - h)^2 - p_j^2 \\ &= 2h(p_i - p_j) + 2h^2. \end{aligned}$$

The $2h^2$ term is a finite-step correction. If the receiver and donor have equal probabilities, the transfer makes them unequal and raises concentration by $2h^2$. If the donor has

exactly one more token than the receiver, the move swaps their counts and leaves concentration unchanged.

This is a useful example of what a discrete derivative retains: the actual size of the move. Replacing it by only its term proportional to h would miss an exact effect at equal coordinates. Later, the local Fisher formula will also be presented alongside an exact finite expression, so that the effect of approximation remains visible.

We can compare fields on the same graph without changing its geometry of allowed moves. A cost field might assign greater weight to one survey response than another, while concentration and entropy treat labels symmetrically. Directional differences depend jointly on the chosen field and the chosen edge.

3.4 An information field for the same lattice

A second scalar field is **entropy**:

$$H(p) = -\sum_i p_i \ln p_i, \qquad 0 \ln 0 := 0.$$

Here $\ln z$ is the natural logarithm: the exponent needed to express z as a power of e. For example, $\ln 1 = 0$ and $\ln(1/2) \approx -0.6931$. A calculator's ln key is enough to evaluate this field. Chapter 5 will explain its meaning and derive its main properties.

For now, entropy measures uncertainty in the category of one randomly selected response. Pure states have entropy zero. Spreading probability more evenly generally gives greater uncertainty.

Example 3.3 (Computing an entropy difference)**.** For $p = (1/2, 1/4, 1/4)$,

$$H(p) = -\frac{1}{2}\ln\frac{1}{2} - 2\left(\frac{1}{4}\ln\frac{1}{4}\right) = \frac{3}{2}\ln 2 \approx 1.0397.$$

After the transfer C to A, the state is $q = (3/4, 1/4, 0)$ and

$$H(q) = -\frac{3}{4}\ln\frac{3}{4} - \frac{1}{4}\ln\frac{1}{4} \approx 0.5623.$$

Thus $D_{A\leftarrow C}H(p) \approx -0.4774$. Concentration increased while uncertainty decreased. The transfer A to B merely permutes the entries and leaves entropy unchanged.

A zero contribution for a zero probability means that an outcome which never occurs contributes nothing to the average. We do not ask the calculator to evaluate $\ln 0$.

3.5 What a discrete gradient records

In an interval, there are two possible directions from an interior point. In a triangle, there are several. A **discrete gradient field** records changes of a scalar field along edges.

For an oriented edge $p \to q$, define

$$A_F(p, q) = F(q) - F(p).$$

Reversing the edge reverses the sign: $A_F(q,p) = -A_F(p,q)$.

Two independent lattice directions are $u = e_A - e_C$ and $v = e_B - e_C$. Where both forward moves are legal, one may record the pair

$$\nabla_d F(p) = (D_u F(p), D_v F(p)).$$

It lists changes in a chosen pair of directions. The directions meet at an angle in the triangle, so this pair is a coordinate record, rather than automatically an ordinary arrow perpendicular to contours. The full edge field is often easier to use because it also works at boundaries.

A simple ascent rule examines every legal neighbor, chooses one with the greatest positive increase in F, and stays put if none improves it. Ties require a fixed choice rule. Because the lattice has finitely many states and every actual move strictly raises F, this rule cannot revisit a state. It must stop after finitely many moves.

The stopping state is a local maximum with respect to the permitted edges. For a general field, a local maximum need not be the largest value anywhere on the graph. For entropy on this complete token-transfer lattice, balancing the counts identifies the maxima; Chapter 5 explains why.

A reproducible ascent rule requires more than the phrase "move uphill." At a state, list its legal neighbors, evaluate the scalar field at each, and compare the increases. Select the largest strictly positive increase. If two increases tie, use a declared ordering of categories or another fixed rule. If none is positive, return the same state.

This algorithm needs no formula for a continuous gradient. On a graph with S states, it can make at most $S - 1$ strict-improvement moves, because each visited field value is greater than its predecessor and no vertex can recur. The argument does not guarantee that the selected path is the shortest route to a maximum. Ascent and shortest-path optimization are different tasks.

Allowing equal-valued moves changes the termination argument. Two adjacent states with the same value can be visited back and forth forever unless a tie rule prevents it. For entropy, permutations of a balanced count vector can provide such plateaus. The requirement of strict improvement is therefore part of the dynamics.

3.6 Path sums reveal the cancellation behind gradients

Suppose a path visits states p_0, p_1, p_2, p_3. Adding gradient edge values gives

$$\begin{aligned} A_F(p_0,p_1) + A_F(p_1,p_2) + A_F(p_2,p_3) &= [F(p_1) - F(p_0)] \\ &\quad + [F(p_2) - F(p_1)] + [F(p_3) - F(p_2)] \\ &= F(p_3) - F(p_0). \end{aligned}$$

The intermediate values cancel just as they did in Chapter 1. Any two paths with the same endpoints therefore have the same sum of gradient differences. Around a closed path, the sum is zero.

For an arbitrary signed edge field, the sum around a closed route is its **circulation**. A gradient has zero circulation around every closed route. A field that assigns one unit in a clockwise direction around a triangular face has circulation three around that face, so it cannot be a gradient of vertex values.

This test has a useful interpretation. If local changes are supposed to arise from one underlying score, following them around a loop must return that score to its starting value. A nonzero loop total reveals that the edge values contain additional directional information.

3.7 Flux and divergence describe a different feature

An edge field may instead describe **flux**, an amount transported in one step. Choose an orientation for each edge. Positive flux follows that arrow; negative flux goes against it. The **outward divergence** at a vertex is outgoing flux minus incoming flux.

For a clockwise unit flow around a triangle, one unit arrives and one unit leaves at every vertex. Every divergence is zero, but the circulation is three. Zero accumulation at a vertex can coexist with sustained circulation through it.

Conversely, a vertex may send mass to several neighbors and receive little. Its divergence is positive, so its stored mass decreases. This is the reason for the minus sign in the continuity equation developed in Chapter 4.

For any chosen collection of vertices, add their divergences. Every edge joining two vertices in the collection contributes once positively and once negatively, so it cancels. Only edges crossing the boundary remain:

$$\boxed{\text{sum of outward divergences} = \text{net flux leaving the collection.}}$$

This is a finite version of a divergence theorem. Its content is exact accounting along edges.

Example 3.4 (Divergence on a small directed flow diagram)**.** Take three vertices U, V, W with net flows two along $U \to V$, one along $V \to W$, and three along $U \to W$. Outward divergence is five at U, minus one at V, and minus four at W. The sum is zero.
Now consider only the collection $\{U, V\}$. Its divergences sum to four. The internal flow of two from U to V cancels in that sum. The surviving outward flows are three from U to W and one from V to W, also totaling four.

This example separates three levels of description. An edge stores a directional transport amount. A vertex divergence records net departure there. A collection of vertices exchanges mass only through its boundary edges. Moving from one level to the next requires addition and cancellation, rather than an approximation.

The graph in this flow example could represent actual locations. The token graph instead has whole probability distributions as its vertices. Both support the same divergence bookkeeping, but a field on probability states and a probability stored at

physical locations should not be confused. The next chapter will explicitly choose locations as vertices and put probability mass on them.

3.8 Application: cyclic motion need not be a gradient

Three competing strategies can favor one another cyclically. Consider the balanced score table

$$M = \begin{pmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix}.$$

At state p, its scores are

$$f_A = p_B - p_C, \quad f_B = p_C - p_A, \quad f_C = p_A - p_B.$$

At the center all three are zero. Away from the center, the advantages encourage circulation rather than simple movement up one coordinate.

Exponential normalization gives $p_i^+ = p_i e^{\eta f_i}/Z$, with $Z = \sum_i p_i e^{\eta f_i}$. At $p = (1/2, 1/3, 1/6)$ the scores are $(1/6, -1/3, 1/6)$. With $\eta = 1$, the next state is approximately $(0.5755, 0.2327, 0.1918)$. The scores must be recomputed before the next step because the composition has changed.

We can examine one exact diagnostic, the product $P = p_A p_B p_C$. Direct cancellation gives $\sum_i f_i = 0$ and $\sum_i p_i f_i = 0$. Multiplying the update equations therefore yields

$$P(p^+) = \frac{P(p)}{Z^3}.$$

The weighted arithmetic-geometric mean inequality says that for positive numbers a_i and nonnegative weights p_i adding to one,

$$\sum_i p_i a_i \geq \prod_i a_i^{p_i}.$$

It generalizes the fact that the average of two positive numbers is at least their geometric mean. For rational weights it follows by repeating numbers in an ordinary arithmetic-geometric mean inequality. General weights follow by approximation.

Apply it to $a_i = e^{\eta f_i}$. We obtain

$$Z \geq e^{\eta \sum_i p_i f_i} = 1.$$

For an interior state other than the center, the scores differ, giving strict inequality. Consequently $P(p^+) < P(p)$. A finite cyclic update decreases this product; it does not conserve it.

The product is largest at the center, where it equals $1/27$. Its decrease describes crossing toward lower product contours. It does not prove that every coordinate

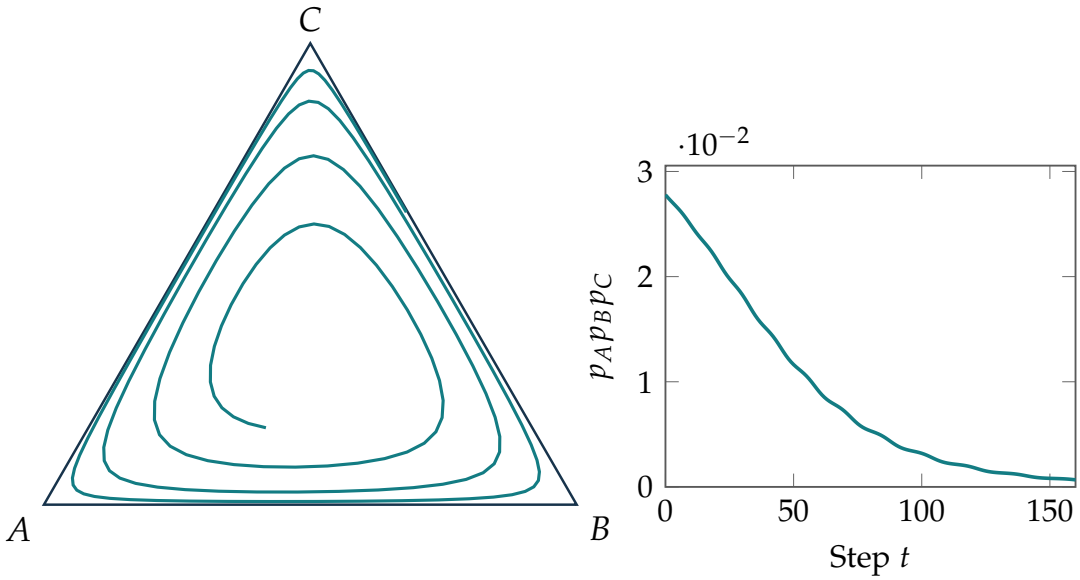


Figure 3.2. Balanced cyclic exponential updates at intensity 0.35. The triangle trajectory circulates while the coordinate product decreases.

approaches zero or that the trajectory has a particular limit. A single scalar diagnostic gives one piece of a multidimensional story.

For comparison, the permutation $T(p_A, p_B, p_C) = (p_C, p_A, p_B)$ has $T^3(p) = p$ and preserves the product exactly. Two rules can both look cyclic while having different invariants. An **invariant** is a quantity unchanged at every step of the particular recurrence.

Use the simplex view of the operator explorer to compare neighboring entropy values. The weekly demonstration also explores lattices and cyclic selection. Chapter 4 turns the flux bookkeeping into a model for transporting actual probability among labeled locations.

3.9 Exercises

3.1 Plot $(1, 0, 0)$, $(0, 1/2, 1/2)$, and $(1/4, 1/4, 1/2)$ in barycentric coordinates.

3.2 List all six states for $N = 2$, and verify the state-count formula.

3.3 List the neighbors of $(2, 1, 1)$ for $N = 4$. Explain why the corner $(4, 0, 0)$ has fewer neighbors.

3.4 For $N = 4$ and counts $(2, 1, 1)$, calculate the concentration differences for transfers A to B and B to C.

3.5 Calculate the entropy difference for the transfer from counts $(2, 1, 1)$ to $(2, 2, 0)$. Compare its sign with the concentration change.

3.6 Explain why $D_{A\leftarrow C}F$ and $D_{C\leftarrow A}F$ at the same starting state need not be negatives of each other. Identify the edge reversal that does give the negative.

3.7 Choose two paths from $(4,0,0)$ to $(2,2,0)$. Add the concentration differences along each path and compare the totals.

3.8 Give a scalar potential on three vertices with values $0, 2, 5$. Calculate its three gradient edge values and the circulation around the triangle.

3.9 Assign a clockwise unit flux around a triangle. Calculate divergence at each vertex and circulation around the triangle.

3.10 Count the strictly positive three-category states for $N = 10$ by first placing one token in each category.

3.11 For the balanced cyclic table, verify $\sum_i f_i = 0$ and $\sum_i p_i f_i = 0$. Derive the product update identity.

3.12 Explain why strict ascent of a scalar field must stop on a finite graph. Describe what could change if equal-valued moves were allowed.

3.10 Selected solutions

2. The states are $(2,0,0), (1,1,0), (1,0,1), (0,2,0), (0,1,1), (0,0,2)$ in counts. The formula gives $(3)(4)/2 = 6$.

3. The neighbors are $(1,2,1), (1,1,2), (3,0,1), (2,0,2), (3,1,0), (2,2,0)$. At the corner only the occupied category can donate.

4. The initial concentration is $3/8$. The first transfer gives $(1,2,1)$ and zero change. The second gives $(2,0,2)$ with concentration $1/2$, hence change $1/8$.

5. The new entropy is $\ln 2 \approx 0.6931$, giving change $-\frac{1}{2}\ln 2 \approx -0.3466$. Concentration rises from $3/8$ to $1/2$, so its change is positive.

8. Orient the triangle $U \to V \to W \to U$. The edge values are $2, 3, -5$. Their circulation is zero.

10. After reserving three tokens, the remaining seven can be distributed in $(8)(9)/2 = 36$ ways. The total number of states, including the boundary, is sixty-six.

Laboratory investigation

1. Take $N = 4$. (a) List all 15 count triples (a, b, c) adding to four. (b) Draw their triangular arrangement or use the lattice demo. (c) Count neighbors of $(2,1,1)$ and $(4,0,0)$. (d) Explain why the boundary has fewer neighbors.
2. Starting at $(2,1,1)$, move one token from C to A and one from A to B, in separate trials. (a) Record both new count triples. (b) Convert the triples to probabilities. (c) Calculate the entropy at each distribution, taking $0 \log 0 = 0$. (d) Determine the signs of both directional differences.

3. At an interior state p, define $u = e_A - e_C$ and $v = e_B - e_C$. (a) Write $p + u/N$ and $p + v/N$. (b) Explain when each transfer is legal. (c) Evaluate the two differences of H at $p = (1/4, 1/4, 1/2)$ with $N = 4$.
4. Choose two different three-step routes on the token lattice with the same start and finish. (a) Label the entropy at each visited point. (b) Add the successive entropy differences along each route. (c) Explain why the route sums agree, even if the intermediate values differ.
5. Draw a triangle of three allowed states and assign a clockwise flow of one unit along each edge. (a) Compute divergence at each vertex. (b) Compute the circulation around the triangle. (c) Explain how zero divergence can coexist with nonzero circulation.